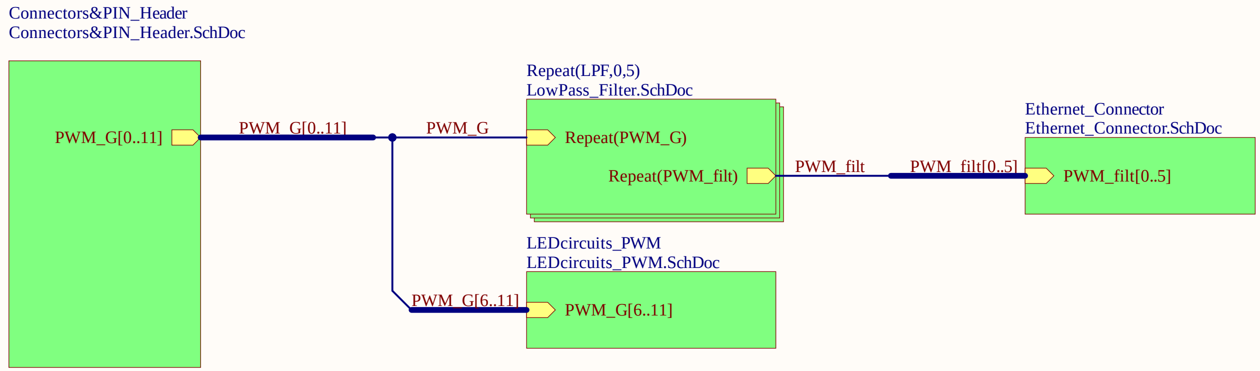


LOGO1
UZ Logo

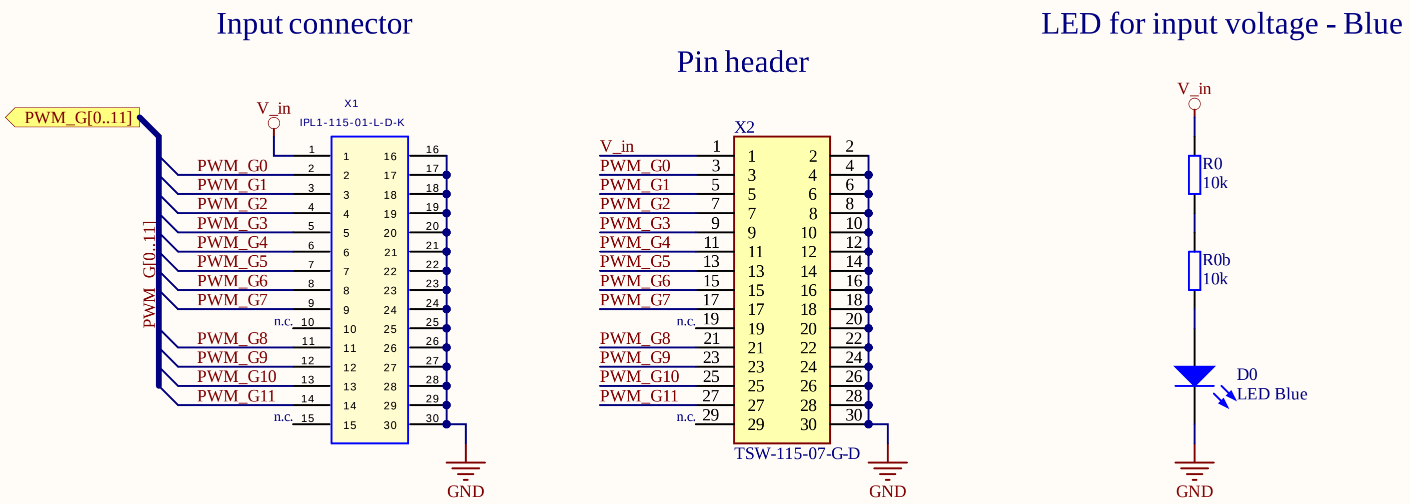
Rev02
Revision

Serial
Serial1

uz_per_loopback_tutorial
Project

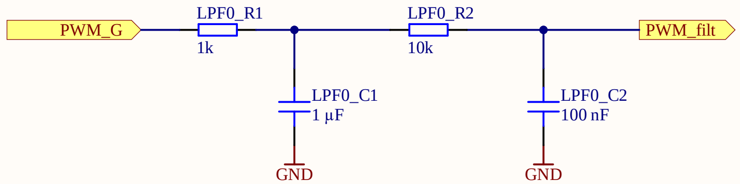
J. Hahn
Designer

Title Interface_TopSheet.SchDoc	**UltraZohm**
Revision: Rev02 | Design Engineer: J. Hahn	www.ultrazohm.com
Project: UZ_PER_loopback_tutorial.PrjPcb	Date: 16.04.2026 Sheet 1 of 10



Title Connectors&PIN_Header.SchDoc	**UltraZohm**
Revision: Rev02 · Design Engineer: J. Hahn	www.ultrazohm.com
Project: UZ_PER_loopback_tutorial.PrjPcb	Date: 16.04.2026 · Sheet 2 of 10

Low pass filters

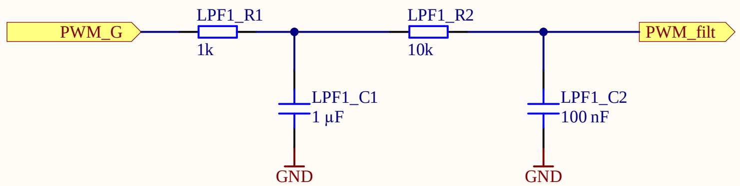


Low pass filter

cutoff-frequency
f_c =
1/(2*pi*sqrt(R1*C1*R2*C2))=
159 Hz

Title LowPass_Filter.SchDoc	**UltraZohm** www.ultrazohm.com
Revision: Rev02 / Design Engineer: J. Hahn	Date: 16.04.2026
Project: UZ_PER_loopback_tutorial.PrjPcb	Sheet 3.0 of 10

Low pass filters

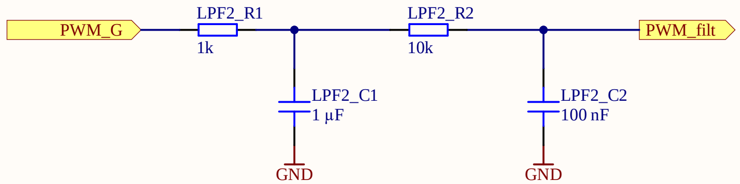


Low pass filter

cutoff-frequency
f_c =
1/(2*pi*sqrt(R1*C1*R2*C2))=
159 Hz

Title LowPass_Filter.SchDoc	**UltraZohm**
Revision: Rev02	Design Engineer: J. Hahn	www.ultrazohm.com
Project: UZ_PER_loopback_tutorial.PrjPcb	Date: 16.04.2026 Sheet 3.1 of 10

Low pass filters

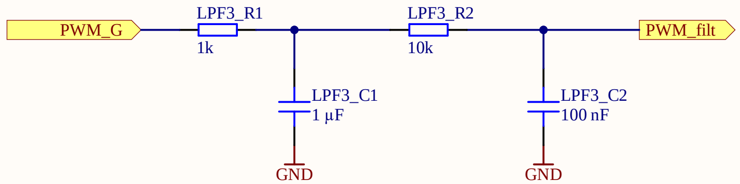


Low pass filter

cutoff-frequency
f_c =
1/(2*pi*sqrt(R1*C1*R2*C2))=
159 Hz

Title LowPass_Filter.SchDoc		***UltraZohm***	
Revision: Rev02	Design Engineer: J. Hahn	www.ultrazohm.com	
Project: UZ_PER_loopback_tutorial.PrjPcb		Date: 16.04.2026	
		Sheet 3.2 of 10	

Low pass filters

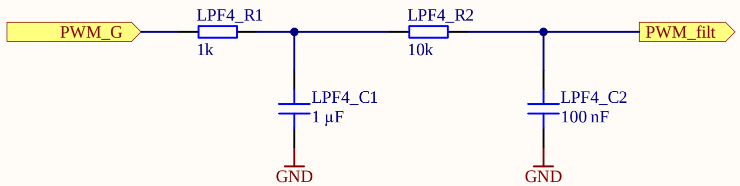


Low pass filter

cutoff-frequency
f_c =
1/(2*pi*sqrt(R1*C1*R2*C2))=
159 Hz

Title LowPass_Filter.SchDoc		**UltraZohm**
Revision: Rev02	Design Engineer: J. Hahn	www.ultrazohm.com
Project: UZ_PER_loopback_tutorial.PrjPcb		Date: 16.04.2026 Sheet 3.3 of 10

Low pass filters

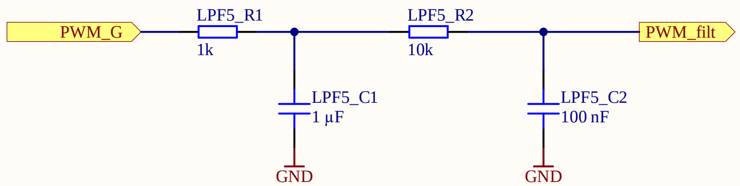


Low pass filter

cutoff-frequency
f_c =
1/(2*pi*sqrt(R1*C1*R2*C2))=
159 Hz

Title LowPass_Filter.SchDoc		**UltraZohm**	
Revision: Rev02	Design Engineer: J. Hahn	www.ultrazohm.com	
Project: UZ_PER_loopback_tutorial.PrjPcb		Date: 16.04.2026 Sheet 3.4 of 10	

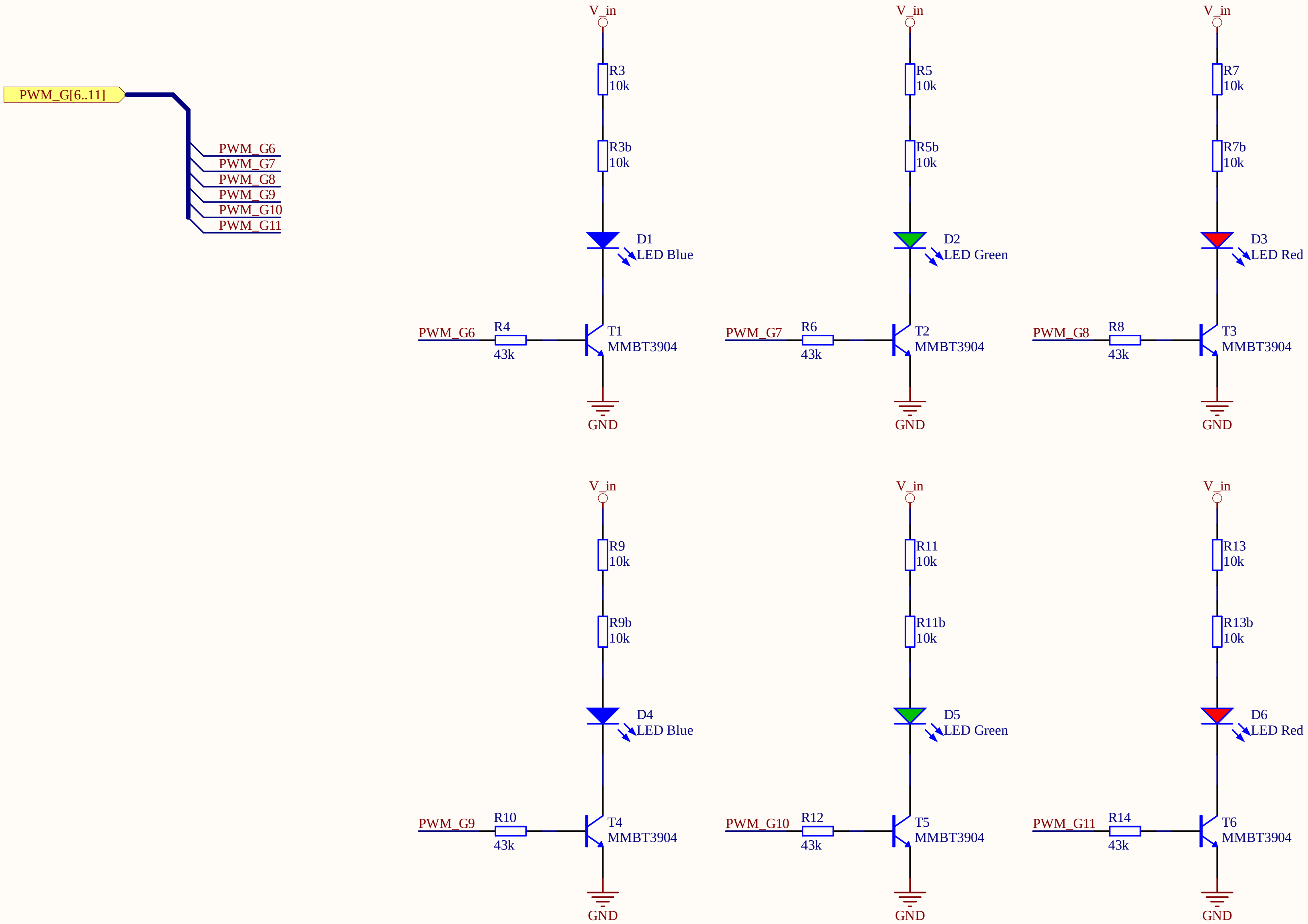
Low pass filters



Low pass filter

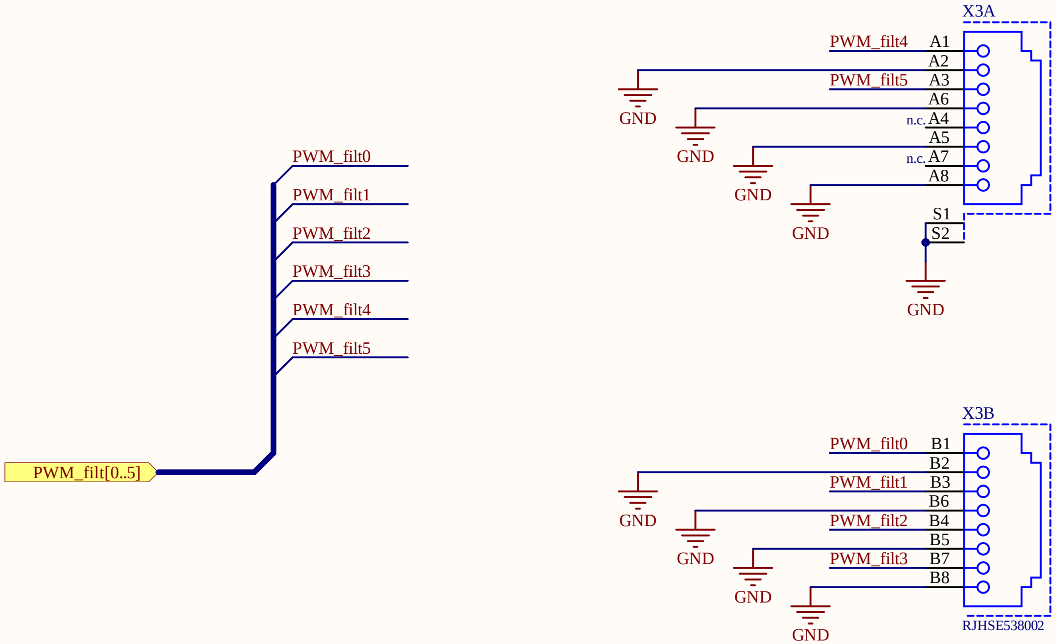
cutoff-frequency
f_c =
1/(2*pi*sqrt(R1*C1*R2*C2))=
159 Hz

Title LowPass_Filter.SchDoc	**UltraZohm**
Revision: Rev02	Design Engineer: J. Hahn	www.ultrazohm.com
Project: UZ_PER_loopback_tutorial.PrjPcb	Date: 16.04.2026 Sheet 3.5 of 10



Title LEDcircuits_PWM.SchDoc		***UltraZohm*** www.ultrazohm.com
Revision: Rev02	Design Engineer: J. Hahn	
Project: UZ_PER_loopback_tutorial.PrjPcb		Date: 16.04.2026 Sheet 4 of 10

Ethernet Connectors



Title Ethernet_Connector.SchDoc

Revision: Rev02 | Design Engineer: J. Hahn

Project: UZ_PER_loopback_tutorial.PrjPcb

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Date: 16.04.2026
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